Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

CompTIA Security+ Exam Prep (SY0-701)

Course Code: C628

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 July 2026	Initial release — Security+ SY0-701 3-day lesson plan aligned to the 36 labs.	Dr. Alfred Ang
2.0	18 August 2026	Reorganised to 5 days aligned to the SY0-701 exam domains (one domain per day); added full concept coverage per exam objective and 7 new labs (43 total); added exam-registration and practice-exam sections.	Dr. Alfred Ang

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Course Information

Course Title	CompTIA Security+ Exam Prep (SY0-701)
Course Reference	C628
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 7.5 instructional hours per day (37.5 hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch; refreshments during activity time)
Mode	Instructor-led, hands-on labs on the free Killercoda Ubuntu Playground
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply general security concepts — control categories and types, the CIA triad and AAA, Zero Trust, change management, cryptographic primitives (hashing, symmetric/asymmetric encryption) and PKI.
- LO2: Identify threats, vulnerabilities and mitigations — threat actors, social engineering, password and web attacks (SQLi, XSS), malware, denial-of-service, wireless attacks and hardening.
- LO3: Implement secure architecture — cloud and enterprise models, network segmentation, firewalls, WAF, VPN, container security, data-at-rest encryption, TLS hardening and resilient backup/RAID.
- LO4: Run security operations — SSH/identity hardening, MFA, SIEM, IDS, EDR, file-integrity monitoring, packet analysis, vulnerability management, incident response, forensics and automation.
- LO5: Manage the security program — governance, risk registers and CVSS, third-party risk, reconnaissance, penetration testing, compliance auditing and security-awareness testing.
- Full attendance is expected — attend all five days to get the most from the course.

Course Schedule

Day 1 — Domain 1 — General Security Concepts (12%)

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, exam-domain overview and ground rules
10:00–11:00	60 min	Domain 1 concepts: security control categories & types, defence in depth, gap analysis

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11:00–12:45	105 min	Domain 1 concepts: CIA/DAD triad, AAA, MFA, Zero Trust, physical security, deception
12:45–13:15	30 min	Lunch break
13:15–14:30	75 min	Domain 1 concepts: cryptography — symmetric/asymmetric, hashing, signatures, PKI, key protection
14:30–16:00	90 min	Hands-on: Lab 1: Security Controls Categorization; Lab 2: CIA Triad and AAA on Linux; Lab 3: Hashing, Salting and Digital Signatures
16:00–17:15	75 min	Hands-on: Lab 4: Symmetric vs. Asymmetric Encryption; Lab 5: PKI and Certificate Authority
17:15–17:30	15 min	Day 1 recap and Q&A

Total instructional time: 450 minutes (7.5 hours).

Day 2 — Domain 2 — Threats, Vulnerabilities & Mitigations (22%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and Q&A
9:45–11:00	75 min	Domain 2 concepts: threat actors & motivations, vectors & attack surfaces
11:00–12:45	105 min	Domain 2 concepts: social engineering, IOCs, password & wireless attacks, application attacks
12:45–13:15	30 min	Lunch break
13:15–15:45	150 min	Hands-on: Lab 6: Phishing Email Analysis; Lab 7: Password Attacks; Lab 8: Malware Static Analysis (Safe Sample); Lab 9: SQL Injection; Lab 10: Cross-Site Scripting
15:45–17:15	90 min	Hands-on: Lab 11: DoS Simulation and Detection; Lab 12: Vulnerability Scanning; Lab 13: Linux Hardening
17:15–17:30	15 min	Day 2 recap and Q&A

Total instructional time: 450 minutes (7.5 hours).

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Day 3 — Domain 3 — Security Architecture (18%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and Q&A
9:45–11:00	75 min	Domain 3 concepts: architecture models, cloud shared responsibility, segmentation, containers
11:00–12:45	105 min	Domain 3 concepts: data protection & states, encryption, DLP, resilience and recovery
12:45–13:15	30 min	Lunch break
13:15–15:45	150 min	Hands-on: Lab 14: Network Segmentation; Lab 15: Firewall with UFW/nftables; Lab 16: Web Application Firewall with ModSecurity; Lab 17: VPN with WireGuard; Lab 18: Disk Encryption with LUKS; Lab 19: TLS Hardening
15:45–17:15	90 min	Hands-on: Lab 20: Backup and Recovery with restic; Lab 37: Cloud Shared Responsibility; Lab 38: Container Security; Lab 39: RAID with mdadm
17:15–17:30	15 min	Day 3 recap and Q&A

Total instructional time: 450 minutes (7.5 hours).

Day 4 — Domain 4 — Security Operations (28%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and Q&A
9:45–11:00	75 min	Domain 4 concepts: secure baselines & hardening, asset management, monitoring and alerting
11:00–12:45	105 min	Hands-on: Lab 21: SSH Hardening + Fail2ban; Lab 22: IAM with PAM and sudo; Lab 23: MFA with TOTP; Lab 24: SIEM with ELK/Wazuh
12:45–13:15	30 min	Lunch break
13:15–14:15	60 min	Domain 4 concepts: vulnerability management & CVSS, IAM/MFA, incident response and

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		forensics
14:15–15:45	90 min	Hands-on: Lab 25: Network IDS with Suricata; Lab 26: Endpoint Detection with osquery; Lab 27: File Integrity with AIDE; Lab 28: Vulnerability Management with Trivy
15:45–17:15	90 min	Hands-on: Lab 29: Incident Response and Forensics; Lab 30: Security Automation and SOAR
17:15–17:30	15 min	Day 4 recap and Q&A

Total instructional time: 450 minutes (7.5 hours).

Day 5 — Domain 5 — Program Management, Oversight & Assessment (20%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 4 recap and Q&A
9:45–11:15	90 min	Domain 5 concepts: governance, policies/standards/procedures, risk management and CVSS
11:15–12:45	90 min	Domain 5 concepts: audits & assessments, penetration testing, third-party risk, security awareness
12:45–13:15	30 min	Lunch break
13:15–15:00	105 min	Hands-on: Lab 31: Risk Register and CVSS Scoring; Lab 32: OSINT Reconnaissance; Lab 33: Active Reconnaissance with Nmap; Lab 34: Penetration Testing with Metasploit (Authorised)
15:00–16:00	60 min	Hands-on: Lab 35: Compliance Audit with OpenSCAP; Lab 36: Phishing Awareness with GoPhish
16:00–16:30	30 min	SY0-701 exam review across all five domains — key concepts, command recall and scenario practice
16:30–17:15	45 min	CompTIA exam registration walkthrough and the Tertiary Infotech

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		practice exam
17:15–17:30	15 min	Course wrap-up, feedback and Q&A

Total instructional time: 450 minutes (7.5 hours).

Lab Reference (aligned to exam domains)

Domain / exam area	Weighting	Labs
Domain 01: General Security Concepts	12%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5
Domain 02: Threats, Vulnerabilities & Mitigations	22%	Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13
Domain 03: Security Architecture	18%	Lab 14, Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20, Lab 37, Lab 38, Lab 39
Domain 04: Security Operations	28%	Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27, Lab 28, Lab 29, Lab 30, Lab 40, Lab 41, Lab 42, Lab 43
Domain 05: Security Program Management & Oversight	20%	Lab 31, Lab 32, Lab 33, Lab 34, Lab 35, Lab 36

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